

Let us finally answer the question of when and in what sense do the Cosine/Sine/Full Fourier series converge to the given function.

- First some definitions and notation : In what follows $a < b$ are real numbers and $f_n(x), f(x)$ defined on $[a, b]$.

NOTIONS OF CONVERGENCE :

DEFINITION 1 : (POINTWISE) : We say that $\sum_{n=1}^{\infty} f_n(x)$ converges to $f(x)$ pointwise in (a, b) if

$$\left| f(x) - \sum_{n=1}^N f_n(x) \right| \rightarrow 0 \text{ as } N \rightarrow \infty \text{ for each } x \text{ in } (a, b).$$

That is,

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N f_n(x) \rightarrow f(x) \text{ for each } x \text{ in } (a, b).$$

Definition 2: We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges UNIFORMLY to $f(x)$ in $[a, b]$ if

$$\max_{a \leq x \leq b} \left| f(x) - \sum_{n=1}^N f_n(x) \right| \longrightarrow 0 \text{ as } N \rightarrow \infty$$

NOTE: ① Endpoints are included in the definition

② One first takes the max over $[a, b]$ and then the limit as $N \rightarrow \infty$.

Definition 3 (L^2 -convergence): We say that the series $\sum_{n=1}^{\infty} f_n(x)$ converges in mean-square (or L^2) sense to $f(x)$ in (a, b) if

$$\int_a^b \left| f(x) - \sum_{n=1}^N f_n(x) \right|^2 dx \longrightarrow 0 \text{ as } N \rightarrow \infty$$

13) Example: $f_n(x) = (1-x)x^n \quad 0 < x < 1 \quad I = (0, 1)$.

$$\sum_{n=1}^N f_n(x) = \sum_{n=1}^N x^{n-1} - x^n = 1 - x^N \xrightarrow[N \rightarrow \infty]{\text{TELESCOPING SUM.}} 1 \quad (x < 1)$$

• So we get pointwise convergence of the series on $(0, 1)$ to $1 = \sum_{n=1}^{\infty} (1-x)x^n$.

• But the convergence is NOT uniform because

$$\max_{0 \leq x \leq 1} [1 - (1-x^N)] = \max_{0 \leq x \leq 1} x^N = 1 \quad \forall N$$

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0.

• Series converges in mean-square sense:

$$\int_0^1 |x^N|^2 dx = \frac{1}{2N+1} \rightarrow 0 \text{ as } N \rightarrow \infty$$

$$\int_0^1 |1 - (1-x^N)|^2 dx$$

In the following we consider sine, cosine or full Fourier series, all of which we denote by $\sum_{n=1}^{\infty} C_n X_n(x)$ where x is in some interval $I = [a, b]$.

We want to know when for a given function $f: I \rightarrow \mathbb{R}$, its associated (sine, cosine or full Fourier) series converges to $f(x)$ and in what sense: pointwise? uniform? square-mean?

THEOREM 1 (UNIFORM CONVERGENCE)

The series $\sum_{n=1}^{\infty} C_n X_n(x)$ converges to $f(x)$ uniformly on $[a, b]$ provided that:

i) f and f' exist and are continuous $\forall a \leq x \leq b$

and

ii) f satisfies the given BC $\left(\begin{array}{l} \text{Dirichlet} \leftrightarrow \text{sine} \\ \text{Neumann} \leftrightarrow \text{cosine} \\ \text{Periodic} \leftrightarrow \text{full Fourier} \end{array} \right)$

Remark: Uniform convergence can be sometimes deduced by the Weierstrass - test :

Let $\{f_j\}_{j=1}^{\infty}$ be a sequence of functions on $[a, b]$ /
 $|f_j(x)| \leq M_j \quad \forall x \in [a, b], j \geq 1$. If $\sum_{j=1}^{\infty} M_j < \infty$
then $\sum_{j=1}^{\infty} f_j(x)$ converges uniformly on $[a, b]$.

Example: Consider $f_j(x) = 2^{-j} \sin(2^j x) \quad j \geq 1$
These functions oscillate erratically as $j \rightarrow \infty$ but $|f_j(x)| \leq 2^{-j}$
and $\sum_{j=1}^{\infty} 2^{-j} < \infty$. So $\sum_{j=1}^{\infty} f_j(x)$ converges $\forall x$
uniformly.

In this case we don't know what $f(x)$ is other than

$$f(x) := \sum_{j=1}^{\infty} 2^{-j} \sin(2^j x)$$

Theorem 2 (L^2 -convergence): The series $\sum_{n=1}^{\infty} C_n X_n(x)$ associated to $f(x)$ converges to $f(x)$ in (a, b) in the mean-square sense provided $\int_a^b |f(x)|^2 dx < \infty$.

Remark: In this case we also have PARSEVAL'S IDENTITY on $[a, b]$:

$$\int_a^b |f(x)|^2 dx = \sum_{n=1}^{\infty} |C_n|^2 \left(\int_a^b |X_n(x)|^2 dx \right)$$

DEFINITION: A function $f(x)$ has a jump discontinuity at $x=c$ if the one-sided limits $f(c^+)$ and $f(c^-)$ exist but are not equal.

The value of the jump discontinuity is $\left[f(c^+) - f(c^-) \right]$

NOTE: value of $f(c)$ if defined is irrelevant.

DEFINITION: A function $f(x)$ is called piecewise continuous on $[a, b]$ if it's continuous except at a finite number of points where f has jump discontinuities.

THEOREM 3 (Pointwise) ① The series $\sum_{n=0}^{\infty} C_n X_n(x)$ associated to $f(x)$ converges pointwise to $f(x)$ for x in (a, b) provided $f(x)$ is continuous on $[a, b]$ and $f(x)$ is piecewise continuous on $[a, b]$.

② If $f(x)$ is itself piecewise continuous on $[a, b]$ and $f'(x)$ is piecewise continuous on $[a, b]$ then the series converges to $\frac{1}{2} [f(x^+) + f(x^-)]$ for all $a < x < b$.

③ If $f(x)$ is a periodic function on $\mathbb{R}$ with period 2ℓ / f and f' are piecewise continuous then the full Fourier series converges to $\frac{1}{2} [f(x^+) + f(x^-)] \forall x \in \mathbb{R}$.

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Example: Let $I = [0, \pi]$ $f(x) = 1$

$\sum_{n \text{ odd}} \frac{4}{n\pi} \sin(nx)$ is the series associated to $f(x) = 1$ on $(0, \pi)$

- The series converges at each $x \in (0, \pi)$ POINTWISE but not uniformly on $[0, \pi]$: one reason is that the series is equal to zero at $x=0$ and at $x=\pi$ while $f(x) = 1$ at $x=0$ and $x=\pi$. So condition i) of Theorem 1 is not satisfied.

The pointwise convergence on $(0, \pi)$ follows from Theorem 3i)

Note that in particular this tells us that

$$\textcircled{+} \quad 1 = f(\pi/2) = \sum_{n \text{ odd}} \frac{4}{n\pi} (-1)^{\frac{(n-1)}{2}} = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$$

- The series converges in L^2 -sense on $(0, \pi)$ since $\int_0^\pi 1^2 dx < \infty$

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In other words

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$$